CHRISTOPHER GONZALES

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EDUCATION

Florida International University **Miami, FL**
Bachelor of Science in Mechanical Engineering - 3.38 GPA *Expected Graduation: May 2028*

- **Concentrations:** Design and Manufacturing
- **Related Coursework:** SolidWorks 3D Modeling, Distinct Mathematics, Programming for Systems, Data Analysis, Circuits

Honors: High Honors, King Solomon Award Winner, Dean's List, **2025 HSF Scholar**

TECHINICAL SKILLS

Engineering: SolidWorks (CSWA/P), Siemens NX, MATLAB, Thermal Analysis, IR Thermography, AutoCAD
Programming & Controls: Python, Java, PHP, JavaScript, GitHub (Version Control); familiar with embedded systems and control logic design
Productivity: Microsoft Excel, PowerPoint, Word (Advanced)
Languages: Bilingual Fluency in English and Spanish.
Key Strengths: Mechanical/electrical system integration, troubleshooting, quick learning of technical systems, project documentation

EXPERIENCE

IEEE Electronics Packaging Society Fellowship **Miami, FL**
Mechanical Engineering Research *September 2025 – January 2026*

- Optimized **thermal management** for electronic components by using high-resolution infrared profiles to identify hotspot formations.
- Validated heat dissipation models through **controlled lab experiments** and delivered documented results that informed cooling strategies for a team of **5 engineers.**

NASA L'Space Mission Concept Academy **Miami, FL**
Systems Engineering Lead *August 2025-December 2025*

- Accomplished project approval by authoring a **50+ page Preliminary Design Review (PDR)** which served as the core technical roadmap for **NASA evaluation**.
- Ensured 100% subsystem integration by collaborating with a multidisciplinary team to sync **mechanical, electrical, and software interfaces** which reduced design conflicts.

Skyline Space Internship **Coimbatore, India**
Mechanical Engineering Intern – UAV and Drones *August 2025-October 2025*

- **Modeled 15+ custom UAV chassis components in SolidWorks** by utilizing 3D assembly tools to ensure precise fitment of rotors, sensors, and flight controllers.
- Collaborated with a **global team of 100+ engineers** via remote platforms to sync design updates and troubleshoot UAV system integration issues.

Presidente Supermarket **Miami, FL**
Inventory & Merchandise Lead *June 2021 - Present*

- Managed **inventory logistics for 200+ product lines**, using organizational systems to ensure 100% stock availability and shelf optimization.
- Reduced **transaction errors by 10%** by auditing price tags and implementing a more efficient sorting system for incoming merchandise.

PROJECTS

ASME 2026 XRC Autonomous Ford Mustang GTD - Developer & Designer **Miami, FL**
(XRC)-Autonomous Vehicles - Lead Developer & Designer *February 2026 – Present*

- Designed a **scale Ford Mustang GTD** on **SolidWorks** to model the aerodynamic body and structural chassis components for competition.
- Programmed vehicle control logic t**o automate drivetrain response and steering**, ensuring high-speed stability and precision handling.

WC Server – Founder & Lead Developer **Miami, FL**
Mojang Studios - Freelance Project *September 2025- Present*

- Architected a high-traffic game server **supporting 100+ concurrent players**, using PHP to develop custom plugins and unique gameplay mechanics.
- Optimized server-side operations by resolving technical bottlenecks and troubleshooting system latency to ensure **99.9%** uptime for the user base.

STRATA Mars Rover – Mission Concept Design Project **Miami, FL**
NASA Professional Workforce Preparation *August 2025– December 2025*

- Developed the **STRATA Mars Rover** concept by performing **mission feasibility analysis and designing system architectures** as part of a multidisciplinary team.
- **Designed the Mars Rover in CAD software** and analyzed location points to determine best optimal placement for mission components and sensors.

LEADERSHIP & ACTIVITIES

Engineering Research Society Engineer – *Technical Coordinator* **Miami, FL**
Florida International University *August 2025 – Present*

- Coordinated **technical workshops for 100+ students**, managing logistics and funding for certification programs in SolidWorks and introductory research methodologies.
- Partnered with **executive leadership** to secure organizational funding and resources, ensuring the successful execution of weekly engineering development seminars.

Panther Motorsports (Formula SAE) – *Mechanical Design Engineer* **Miami, FL**
Florida International University *September 2025 – Present*

- Worked with a 50+ member multidisciplinary team to test **vehicle performance, analyze data, and recommend mechanical improvements** for efficiency and handling.
- Developed **high-fidelity 3D CAD models** for vehicle sub-assemblies in SolidWorks to ensure all components maintained strict clearance tolerances and integrated seamlessly with the chassis.